| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 07/01/2015 |
| **PI Name:** | Rubin, David M.S. | | |
| **Project Title:** | Middleware for Exploration Medical System | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Operational and clinical research | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **ExMC**:Exploration Medical Capabilities | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | david.a.rubin@nasa.gov | **Fax:** | FY |
| **PI Organization Type:** | NASA CENTER | **Phone:** | 281.461.2728 |
| **Organization Name:** | Wyle Laboratories, Inc./NASA Johnson Space Center | | |
| **PI Address 1:** | Project Engineer/ISS Biomedical Flight Controller | | |
| **PI Address 2:** | 1290 Hercules Avenue | | |
| **PI Web Page:** | | | |
| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 07/01/2011 | **End Date:** | 03/27/2015 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Watkins, Sharmi1a | **Contact Phone:** | 281.483.0395 |
| **Contact Email:** | sharmila.watkins@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 3/27/2015 (from 9/30/2018) per PI (Ed., 6/30/15) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | Directed Research | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

The middleware will be intelligent data management and communications software that facilitates the acquisition, integration, management, and distribution of data from on-orbit vehicle medical and environmental devices, sensors, and systems. The software will reduce crew time and ensure the accuracy of data collection and timeliness of data delivery.
Development of the middleware addresses the following risk as stated within the Critical Path Roadmap (CPR; also known as the NASA Bioastronautics Roadmap (NASA/SP-2004-6113)):

Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities: Given that medical capabilities will be limited during human exploration missions, there is a possibility that in-flight medical events will lead to unacceptable health and mission outcomes.

| | |
|---|---|
| **Task Description:** | The objectives of the Middleware are to:<br><br>a. Reduce or eliminate the time required of an on-orbit crew and ground personnel [which include Surgeon, Biomedical Engineer (BME), and Medical Operations Data Specialist] to access, transfer, and manipulate medical data.<br><br>b. Support the integrated demonstration that the on-orbit crew has the ability to access medical data/information via an intuitive and crew-friendly solution to assist/aid in the treatment of a medical condition.<br><br>c. Develop a common data management framework and architecture that can be ubiquitously used to automate repetitive data collection, management, and communications tasks for all crew health and life sciences activities.<br><br>d. Ensure crew access to medical data during periods of reduced or unavailable ground communication.<br><br>e. Develop a common data management framework and architecture that allows for scalability, extensibility, and interoperability of data sources and data users.<br><br>f. Lower total cost of ownership for development and sustainment of peripheral hardware and software that utilize the Middleware for data management<br><br>g. Provide better crew health via reduction in crew errors, crew time, and ground time.<br><br>The ground demonstration of the Middleware, as part of the Exploration Medical System Demonstration (EMSD), will occur in 2015. |
| **Rationale for HRP Directed Research:** | This study is a combination of a solicited task (RFP released under Wyle for procurement of COTS middleware system) and a directed task involving customization of the software for the specific needs of ExMC. The study team is uniquely positioned to perform this function and to ensure that the middleware component of the overall Exploration Medical System Demonstration (EMSD) is integrated with the other components of the system. |
| **Research Impact/Earth Benefits:** | The Exploration Medical System is being designed to demonstrate the integration of complimentary but, until recently, disparate pieces of medical informatics software. The Middleware software, as part of the larger Exploration Medical System, has potential training and simulation opportunities for future astronauts and ground personnel. In addition, the Exploration Medical System can serve as software infrastructure for future development and integration of medical devices and software. |
| **Task Progress:** | Task progress made during the reporting period includes the development, integration, and testing of all EMSD components (i.e., electronic medical record, assisted medical procedures software, Middleware infrastructure, associated Middleware "adapters").<br>The integrated EMSD system was demonstrated in the Human Research Program (HRP) Human Exploration Research Analog (HERA) environment for project stakeholders.<br><br>Project documentation was completed and submitted to the proper repositories. |
| **Bibliography Type:** | Description: (Last Updated: 04/16/2014) |